

Broadening Topic Breadth by Hiding the Latest Message in Text Chat

Asuka Ishii

The Department of Media Creation
Institute of Advanced Media Arts and Sciences
Ogaki, Gifu, Japan
asuka25@iamas.ac.jp

Kensuke Tobitani

Department of Media Creation
The Institute of Advanced Media Arts and Sciences
Ogaki, Gifu, Japan
tobitani@iamas.ac.jp

Abstract

Text-based computer-mediated communication (CMC) interfaces foreground the most recent message, promoting rapid replies but potentially discouraging exploratory dialogue. We introduce Masked Chat, a chat interface that temporarily conceals the most recent message to create an intentional information gap. In an experiment with previously unacquainted dyads, the masked condition increased perceived topic breadth and yielded marginally higher lexical diversity than a baseline interface, without increasing subjective cognitive load. These findings suggest that selectively limiting visibility can prompt contextual inference and active meaning-making, shifting users away from purely reactive responding. We propose Richness through Hiding as a practical design lever for supporting exploratory dialogue in efficiency-oriented text chat.

CCS Concepts

• **Human-centered computing** → *Social engineering (social sciences); Natural language interfaces; Text input.*

Keywords

Computer Mediated Communication, Interaction Design, Text Entry, Text/Speech/Language, Lab Study, Quantitative Methods

ACM Reference Format:

Asuka Ishii and Kensuke Tobitani. 2026. Broadening Topic Breadth by Hiding the Latest Message in Text Chat. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798600>

1 Introduction

In HCI and interface design, increasing information visibility has long been a central strategy for supporting smooth interaction [7, 11]. In text-based computer-mediated communication (CMC), interface features such as conversation history[9], read receipts[3], and typing indicators[20] are widely assumed to facilitate coordination and conversational flow. However, does making all information visible always improve conversational quality? High visibility can

reduce the room for interpretation—the space in which users actively infer context and fill informational gaps[1]—and may thereby shift dialogue toward reactive information exchange.

In CMC, interactional phenomena such as disrupted adjacency (i.e., turns that no longer align with the immediately prior message) and overlapping exchanges (i.e., multiple topical threads progressing in parallel) have been documented [16, 17, 24, 29]. Although these patterns are sometimes treated as user confusion, an alternative interpretation is that breakdowns emerge because chronological interfaces enforce linear adjacency even when participants organize meaning across non-linear temporal relationships. This interpretation highlights a contrast with face-to-face conversation, where people can maintain coherence while managing multiple topics and flexibly returning to earlier context [6, 28]. Together, these observations suggest that linear conversational patterns in CMC are not solely a consequence of human cognitive limitations but may also be amplified—or even imposed—by interface constraints.

Gaver et al. argue that ambiguity can serve as a design resource by prompting interpretation and supporting meaning-making[10]. However, empirical work applying this idea to text chat—where efficiency is often paramount—remains limited, and it is still unclear whether intentional information masking reliably encourages interpretive engagement and meaning-making.

If the HCI community continues to assume that increased visibility inherently benefits users, CMC may remain optimized primarily for efficient information transmission, potentially eroding opportunities to explore context and fill gaps under uncertainty[25–27]. In text chat, chronological layouts make the most recent message salient, biasing attention toward it and increasing the likelihood that users prioritize local coherence with the immediately preceding message when selecting their next action [16, 24]. This bias may suppress exploratory behaviors, such as revisiting earlier context or making opportunistic topic shifts[18]. This study examines whether deliberately concealing a key cue—the content of the most recent message—can interrupt automated response patterns and create opportunities for users to actively explore context and construct meaning. However, intentionally introducing an information gap forces users to rely heavily on memory and contextual inference, which inherently introduces uncertainty and may lead to increased cognitive load. Therefore, we investigate whether this deliberate inconvenience becomes excessive mental burden, outweighing its benefits.

1.1 Hypotheses

This study employs Masked Chat, an experimental chat interface that temporarily conceals the content of the most recent message,



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CHI EA '26, Barcelona, Spain

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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798600>

to examine how intentional information concealment affects conversation. We test the following hypotheses.

H1a. The intervention increases perceived topic breadth.

H1b. The intervention increases lexical diversity.

H2. The intervention increases participants' subjective cognitive load.

By testing these hypotheses, we aim to clarify whether information masking functions not only as an impediment but also as a design resource that elicits exploratory dialogue, using both subjective measures and behavioral metrics.

1.2 Contributions

This paper provides empirical evidence of a trade-off in text-based CMC design between maximizing message visibility and fostering interactional richness, and it derives design implications from this evidence.

First, we show that introducing temporary and intentional information gaps—by concealing the most recent message—can elicit active engagement and exploratory behavior rather than functioning merely as an inconvenience. In doing so, we operationalize and extend prior discussions of the benefit of inconvenience and ambiguity as a resource to a concrete intervention for text-based chat interfaces.

Second, we propose a practical design lever for CMC practitioners who often prioritize efficiency: perceived topic breadth and lexical diversity may be tuned by manipulating recency cues (e.g., most-recent-message visibility). This implication suggests new interface directions that support not only efficiency-oriented communication (e.g., work-related coordination) but also contexts that value divergent thinking and playfulness (e.g., casual chatting and brainstorming).

2 Method

2.1 Apparatus

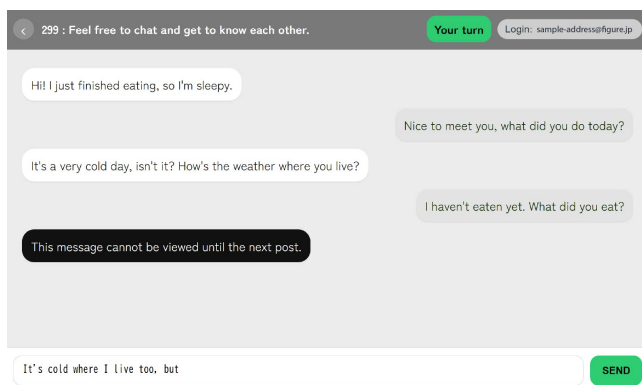


Figure 1: Screenshot of Masked Chat from User A's perspective. After User A sent "I haven't eaten yet. What did you eat?", User B sent a new message. User B's new message is instantly concealed and replaced with the notice, "This message cannot be viewed until the next post". User A must send their next message to reveal User B's hidden text.

As an experimental manipulation, we implemented a web-based chat interface called Masked Chat, which temporarily conceals the content of the most recent message. As shown in Figure 1, the masking and unmasking are precisely triggered by the users' posting actions. When the partner's latest message arrives, it is immediately masked and replaced with a placeholder stating, "This message cannot be viewed until your next post." The mask is removed, and the original text is revealed, at the moment the user sends their subsequent reply. Thus, the recipient can see that a message was sent but cannot access its content until they send the next message. When a participant's own message is the most recent, it remains visible to the sender but is concealed on the recipient's screen. For comparison, we also implemented Baseline Chat, which is identical to Masked Chat except that the masking feature is disabled. Both interfaces ran in a desktop web browser on a PC.

To reduce visual associations with existing chat applications, we used a grayscale palette, with the exception of the send button and the turn-taking indicator described below. Prior HCI work suggests that interface color can influence how users interpret the affective tone of chat messages [23]. Moreover, visual elements resembling well-known platforms may prime interaction styles associated with those platforms [4]. To minimize such priming effects—stemming from prior experience with specific applications and individual social media habits—we adopted a visually neutral design.

Prior work also suggests that imbalanced turn-taking can influence topic dominance, potentially biasing the discussion toward the interests of the more active participant [19]. To reduce this confound and strictly isolate the effect of the masking intervention from interactional dominance, both Masked Chat and Baseline Chat enforced alternating turns (one message per turn). Specifically, when it is a user's turn, "Your turn" appears in the top-right corner. Upon sending a message, the turn transitions to the partner, and the display changes to "The other person's turn." In this state, the send button is disabled, mechanically preventing multiple consecutive posts by the same user. The only functional difference between the two interfaces was the presence or absence of message masking, enabling us to examine how concealing the most recent message shapes perceived topic breadth and the conversational strategies participants adopt in response to the intervention.

2.2 Participants

We recruited Japanese-speaking participants aged 20–29 who use text-based chat on a daily basis and are comfortable with keyboard input. We restricted the age range to the twenties because large age differences within a pair can introduce age-related deference and conversational avoidance [12], which may confound natural topic development. Participants were recruited via the experiment participant recruitment site (<https://www.jikken-baito.com>), resulting in 10 males and 10 females, for a total of 20 participants. The average age of participants was 24.235 years, with a standard deviation of 2.796. The youngest participant was 20, and the oldest was 29.

2.3 Ethics

This study was approved by the Research Ethics Review Committee at the authors' affiliated institution (Approval No. 122). Before participation, all participants were informed about the study purpose

and procedures, the anonymization of their data, and their right to withdraw at any time without penalty. Informed consent was obtained via Google Forms. All data were anonymized, securely stored, and analyzed in a manner that prevented identification of individual participants.

2.4 Procedure

Two previously unacquainted participants were paired and engaged in a 20-minute text-chat conversation using either Masked Chat or Baseline Chat. Each participant completed both conditions; however, they were paired with a different partner (also a stranger) in each condition. Condition order (Masked→Baseline vs. Baseline→Masked) was counterbalanced across participants. Because women tend to report slightly higher self-disclosure than men [5], and self-disclosure in stranger conversations may affect perceived topic breadth, we used blocked pairing to balance gender composition across conditions and reduce confounding due to gender pairing. As a result, each condition included 10 pairs: 4 mixed-gender pairs, 3 male-male pairs, and 3 female-female pairs.

Before the session, participants watched an instructional video explaining the interface. They then completed a 3-minute practice chat with an AI agent to familiarize themselves with the UI. Next, they conducted a 20-minute conversation with their assigned partner. The interface displayed the following prompt: “Feel free to chat and get to know each other. For example, school, work, free time, today’s activities, dreams, likes, and dislikes could be topics. Please avoid discussing the experimenter or the system used in this study.” Participants completed questionnaires before the experiment and after each trial. After completing both sessions, participants received a JPY 3,000 Amazon gift card as compensation.

2.5 Measures

Participants completed questionnaires before the experiment and after each trial.

Before the experiment, participants reported their age and gender and completed the Big Five personality inventory [2, 13] to assess personality traits. Based on prior research [8], we expected that a larger difference in Openness between interlocutors would lead the more open interlocutor to introduce and explore new topics with the less open interlocutor. Therefore, we collected Big Five data to include Openness difference as a covariate in the analysis.

After each trial, participants completed a perceived topic breadth measure (“The breadth of topics we discussed,” 7-point semantic differential scale) corresponding to Hypothesis 1a and the NASA-TLX (Raw TLX) [14, 15] corresponding to Hypothesis 2 to assess cognitive load.

For Hypothesis 1b, we operationalized lexical diversity as the number of unique morpheme types observed in each chat trial [21]. We conducted morphological analysis using McCab[22] and UniDic [30], and retained only nouns, verbs, adjectives, and adverbs. Following the implementation in statsmodels, we used Wald z-tests for fixed-effect inference.

3 Analysis

Data analysis was conducted in Python using the statsmodels library. The significance threshold was set to $\alpha = 0.05$, and results with $p < .10$ are reported as marginal trends.

3.1 Individual-level outcomes (H1a, H2)

For H1a (perceived topic breadth) and H2 (cognitive load), we fit linear mixed-effects models (LMMs). Condition (Masked vs. Baseline) was included as a fixed effect, and participant ID was included as a random intercept. This modeling approach allows the use of all available observations under missingness (e.g., when a participant completed only one condition). For H1a, we additionally included the absolute difference in TIPI Openness within each pair as a covariate, and tested its interaction with Condition.

3.2 Group-level outcome (H1b)

For H1b (lexical diversity), we used the pair (i.e., trial) as the unit of analysis. Because the sample size was insufficient to support a fully multilevel model that accounts for non-independence due to participant overlap, we treated each pair as an independent observation and fit an ordinary least squares (OLS) regression model with Condition as the predictor. To avoid overfitting, we did not include the absolute TIPI Openness-difference covariate in this model.

4 Result

4.1 H1: Topic Diversification

For H1a (perceived topic breadth), a linear mixed-effects model showed a significant main effect of Condition ($< .001^{***}$). Participants in the Masked condition reported greater perceived topic breadth than those in the Baseline condition (Masked: $M = 4.33$, $SD = 1.08$; Baseline: $M = 3.22$, $SD = 1.16$). We also observed marginal trends for the main effect of absolute Openness difference ($p = 0.73$) and for the Condition $\times$ Openness-difference interaction ($p = 0.65$).

For H1b (lexical diversity), OLS regression analysis showed a marginally significant main effect of condition ($p = 0.82$). The Masked condition yielded higher lexical diversity (i.e., more unique morphemes) than the Baseline condition (Masked: $M = 122.00$, $SD = 35.39$; Baseline: $M = 95.11$, $SD = 25.17$).

4.2 H2: Cognitive Load

For H2 (cognitive load), we found no statistically significant difference between conditions ($< .001^{***}$). Cognitive load scores did not differ reliably between the Masked condition ($M = 30.09$, $SD = 11.19$) and the Baseline condition ($M = 26.25$, $SD = 13.08$).

5 Discussion

The results indicate that concealing the most recent message broadened perceived topic breadth (supporting H1a). Regarding lexical diversity, the results showed only a marginal tendency ($p < .10$); thus, we position this quantitative finding as a preliminary trend rather than definitive evidence of active contextual exploration (partially supporting H1b). Furthermore, we found no significant increase in subjective cognitive load (not supporting H2). This pattern

Table 1: Results of Linear Mixed Models (H1a, H2) and Linear Regression (H1b). Significance levels: * $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.1$.**

Hypothesis & Dependent Variable	Effect / Parameter	Coef.	SE	Test Stat. (t/z)	p -value
<i>Individual-level Analysis (Linear Mixed Model, $N_{obs} = 36^*$)</i>					
H1a: Perceived Breadth of Topics (1-7 scale, SD)	Intercept	2.70	0.38	7.04	< .001***
	Condition [Masked]	1.90	0.56	3.36	< .001***
	Abs Diff in Openness	0.41	0.23	1.79	.073 [†]
	Condition $\times$ Abs Diff in Openness	-0.56	0.30	-1.85	.065 [†]
	Random Effect Var. (Participant)	0.09			
H2: Cognitive Load (0-100 scale, SD)	Intercept	26.07	2.77	9.42	< .001***
	Condition [Masked]	4.17	3.04	1.37	0.169
	Random Effect Var. (Participant)	58.30			
<i>Group-level Analysis (Linear Regression, $N_{pair} = 18$)</i>					
H1b: Lexical Diversity (Unique Morphemes)	Intercept	95.11	10.24	9.29	< .001***
	Condition [Masked]	26.89	14.48	1.86	.082 [†]

Note: Condition was coded as Control = 0, Masked = 1. Test statistics are z -values for Linear Mixed Models and t -values for Linear Regression. N_{obs} indicates the total number of observations included in the mixed models.

challenges the assumption that maximizing information visibility necessarily guarantees interactional richness; instead, introducing an information gap appears to shift conversational behavior from reactive responding to active exploration.

Anecdotal observations from the chat transcripts support this interpretation. For example, in the Baseline condition, participants tended to reply linearly to the immediate preceding question (e.g., Q: “Do you like reading? I was reading a book right before the experiment,” A: “I like it too! I especially like mysteries.”). In contrast, in the Masked condition, participants frequently introduced new parallel topics or elaborated on their own previous statements while waiting to see the partner’s masked message, which likely contributed to the increased perceived topic breadth.

This interpretation aligns with Gaver et al.’s argument that ambiguity can be a design resource that invites multiple interpretations and encourages close personal engagement with a system[10]. By selectively withholding a highly salient recency cue (the most recent message), Masked Chat likely reduced participants’ ability to rely on shortest-path, locally contingent replies and instead encouraged them to reconstruct context through inference and memory. Importantly, the absence of increased subjective cognitive load suggests that this interpretive work may have been experienced as productive engagement rather than as a burden. Furthermore, one possible explanation for this is that concealing the immediate context liberated participants from the cognitive demand of closely aligning their responses with their partner’s preceding message. In typical chat interfaces, users often engage in passive, consideration-heavy tasks to continuously adapt their topics and tone to the immediate context. By relinquishing this need for meticulous coordination, participants may have shifted to an active, self-directed conversational mode, freely introducing topics of their own interest. While this remains speculative, it suggests that the freedom to converse without strict contextual constraints may offset the mental burden introduced by the information gap, consistent with the idea that not all uncertainty functions as friction.

Together, these findings suggest that “richness” in text chat can be supported not only by adding more cues, but also by intentionally limiting visibility to create interpretive space. This supports *Richness through Hiding* as a practical design lever for fostering exploratory dialogue in efficiency-oriented CMC.

6 Conclusion

This study suggests that deliberately concealing the most recent message can promote exploratory dialogue in text-based CMC, increasing perceived topic breadth and lexical diversity without increasing subjective cognitive load. These findings extend prior discussions of the benefit of inconvenience and intentional ambiguity by demonstrating their applicability to efficiency-oriented chat interfaces. We introduce the design principle of *Richness through Hiding*, which motivates reconsideration of the prevailing assumption that maximizing immediate visibility is always optimal for conversational interaction.

6.1 Limitations and Future Work

This experiment focused on short-term conversations between previously unacquainted pairs. However, this design cannot be directly applied to situations demanding accuracy and accountability, such as high-stakes workplace decision-making. Therefore, future work should more precisely characterize when information masking increases the risk of misunderstanding versus when it promotes productive reinterpretation, demonstrating in which contexts information-concealing design strategies prove effective. Given our small sample size ($n = 19$), the quantitative results alone are insufficient to conclusively demonstrate that users actively explored contexts and constructed meanings. Therefore, future studies should incorporate qualitative coding of the chat transcripts to rigorously verify whether and how users engage in active meaning-making. Finally, while we enforced strict turn-taking to isolate the effect of masking, real-world text chat is inherently spontaneous and characterized by overlapping exchanges. It is possible that the observed increase in topic breadth was partially influenced by

the enforced waiting period rather than the masking alone. Future research must adapt the Masked Chat design to non-turn-based contexts, exploring how the temporary masking of messages operates when multiple overlapping threads co-occur naturally.

7 Usage of Generative AI

We utilized a large language model, OpenAI GPT-5.2, to copy-edit and improve the clarity of the English text in this manuscript. The authors wrote the original drafts, carefully reviewed all AI-edited text, and take full responsibility for the final content. Additionally, we employed a generative AI conversational agent (OpenAI GPT-4o API) during the 3-minute practice session of our experiment.

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